

I am building a research platform that manages and extracts insights from large volumes of unstructured data (research papers, reports, filings, PDFs in mixed formats) on companies and industries.

The goal is to build a research platform that allows to efficiently organise and explore these documents, extract insights, make dashboards and answer questions around it.

First of all , the tech stack I will use :

- **FRONTEND - > REACT(VITE) AND TAILWIND CSS**

WHY ?

Because , React with Vite has been considering the fastest frontend framework combo, which has modern styling commands . Vite offers rapid UI development and debugging , and we can use the Material UI library for creating multiple Component of our project such as tables , filters and charts for better responsiveness .

- **BACKEND - > EXPRESS JS WITH MYSQL DATABASE**

WHY ?

EXPRESS JS

Due to its lightweight and flexible nature , it will be the perfect suit for the RESTful API . It allows clear segregation of routes , services and middle wares , which will help to maintain scalable backend .

MYSQL

It helps maintain data consistency and relationships between entities like companies, documents, and industries. It also makes it easier to perform analytical queries that are useful for dashboards and reporting .

I will segregate each and every section based on research paper and files and other documents .

I will make the corresponding UI for showing the data to the user .

For Authentication I will use the JWT and O AUTH . And I will maintain middle wares and store sessions in cookie

- **DEPLOYMENTS - >AWS (EC2, S3) WITH CICD**

For deployments, I would use AWS EC2 for backend services and AWS S3 for document storage, along with a CI/CD pipeline to automate builds, testing, and deployments. This setup ensures faster iteration, fewer manual errors, and consistent releases across environments.

Using CI/CD is especially important in a research platform where features evolve quickly and document processing logic may change frequently.

- **AI FEATURE -> RAG AGENT WITH GEMINI AI WITH VECTOR DB**

I will use the rag agent because it is best in terms of research platform where accuracy, traceability, and up-to-date information matter more than generic responses.

Instead of relying only on a large language model's internal knowledge, the RAG approach allows the model to retrieve relevant information from the platform's document store and generate answers grounded in actual data.

LLM – Gemini AI

- *It is cost-effective compared to many alternatives*
- *It provides fast response times*
- *It performs well for question answering and summarization tasks*
- *It integrates smoothly with backend services*
- *Vector Database – Qdrant -> For semantic search and retrieval*